

TMD Template

Day 1 handout. The 5-section Technical Modeling Document (TMD) skeleton your quad builds across Week 5 — the sibling to your PRD and TCD, one level deeper than the TCD. Print it, pin it, fill it section by section. Each section carries a [Day X] tag telling you when it gets written. Friday it ships, peer-reviewed.

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# Technical Modeling Document – <feature>
**Sibling to:** PRD <link>, TCD <link> |
**Authors:** <quad> | **Status:** Draft / Reviewed

## 1. Data model
[Day 1]
Entity-level model: tables / collections, keys,
relationships, indexes.
Storage choices and the trade-offs that drove them.

## 2. Cloud topology
[Day 2]
Region / AZ choice, managed services, multi-tenancy
stance.
Rough-order-of-magnitude (ROM) cost note.

## 3. API contract
[Day 3]
Resource design, methods, versioning, idempotency,
error semantics.
REST or SOAP – a defended choice.

## 4. System sequence model
[Day 4]
End-to-end happy-path sequence diagram + 1–2
sad/weird-path sequences.
Names every hop, protocol, and failure handler. Each
sequence traces
back to a PRD acceptance criterion.

## 5. Performance baselines & monitoring
[Day 5]
– Baselines: what we measure today (or expect at
launch)
– Targets: pulled from TCD Section 4 SLOs
– Monitoring plan: dashboards, alerts, on-call
signal
– AI-summary validation log: this week's AI-assisted
summaries with
  their validation status (correct / partly correct
  / wrong + evidence)

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How the sections build on each other

- **§1 data model** is the vocabulary the rest of the TMD speaks in — entities named here recur in the API (§3) and the sequences (§4).

- **§3 API contract** is the surface **§4 sequences** exercise; every hop in a sequence is a call defined in §3.
- **§4 sequences** must satisfy the PRD acceptance criteria and stay inside the **§5 / TCD SLO** latency budget — if the sequence sums past the SLO, that's a finding.
- **§5 monitoring** makes the TCD's SLOs *observable* — an SLO with no alert in §5 is an unverifiable promise.

The bar: a TMD an engineer could start building from. It ships alongside the PRD and TCD; together they carry into Week 6's stakeholder negotiation.

Friday review criteria are on the separate `friday-tmd-rubric` handout.